Instructor Information

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Strings

The string data type is used to store string constants. They are delimited by double quotes.
string a = "Hello";

String Concatenation

The concatenation operator (+) can combine strings together. It also has an accompanying assignment operator (+=), which appends a string to another and creates a new string.
string b = a + " World"; // Hello World
a += " World"; // Hello World
When one of the operands is not of a string type, the concatenation operator will implicitly convert the non-string type into a string, making the following assignment valid.
int i = 1;
string c = i + " is " + 1; // 1 is 1
The string conversion is performed implicitly using the ToString method. All types in .NET have this method, which provides a string representation of a variable or expression. As seen in the next example, the string conversion can also be made explicitly.
string d = i.ToString() + " is " + 1.ToString(); // 1 is 1

C# has its own keyword and associated syntax to make it particularly easy to manipulate strings using this class. You can concatenate strings using operator overloads:
string message1 = "Hello"; // returns "Hello"
message1 += ", There"; // returns "Hello, There"
string message2 = message1 + "!"; // returns "Hello, There!"
C# also allows extraction of a particular character using an indexer like syntax:
string message = "Hello"; char char4 = message[4]; // returns 'o'.
Note the string is zero-indexed This enables you to perform such common tasks as replacing characters, removing whitespace, and changing case. The following table introduces the key methods.

String Members

The string type is an alias for the String class. As such, it provides a multitude of methods related to strings, for example, methods like Replace, Insert, and Remove. An important thing to note is that there are no methods for changing a string. Methods that appear to modify a string actually always return a completely new string. This is because the String class is immutable. The content of a string variable cannot be changed unless the whole string is replaced.

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```
string a = "String";
string b = a.Replace("i", "o"); // Strong
b = a.Insert(0, "My "); // My String
b = a.Remove(0, 3); // ing
b = a.Substring(0, 3); // Str
b = a.ToUpper(); // STRING
int i = a.Length; // 6
```

Escape Characters

A statement can be broken up across multiple lines, but a string constant must be on a single line. In order to divide it, the string constant has to first be split up using the concatenation operator.

```
string myString = "Hello " + "World";
```

To add new lines into the string itself, the escape character (\n) is used.

```
string myString = "Hello\nWorld";
```

This backslash notation is used to write special characters, such as a backslash or double quote. Among the special characters is also a Unicode character notation for writing any character.

Character	Meaning	Character	Meaning
\n	Newline	\f	Form feed
\t	Horizontal tab	\a	Alert sound
\v	Vertical tab	\'	Single quote
\b	Backspace	\"	Double quote
\r	Carriage return	\\	Backslash
\0	Null character	\uFFFF	Unicode character (four-digit hex number)

String Compare

The way to compare two strings is simply by using the equal to operator (==).

```
string greeting = "Hi";
bool b = (greeting == "Hi"); // true
```

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Trim String

In C#, you can trim strings using the built-in Trim(), TrimStart(), and TrimEnd() methods. These methods return a **new string** with the modifications, rather than changing the original.

Methods for Trimming Strings

Method Name	Use
String.Trim()	Removes leading and trailing white-space characters, or a specified set of characters, from the current string.
String.TrimStart()	Removes characters from the **beginning** (leading) of a string.
String.TrimEnd()	Removes characters from the **end** (trailing) of a string.

Examples

1. Trim leading and trailing whitespace

This is the most common use, and requires no arguments:

```
string text = "   Hello World!   ";

string trimmedText = text.Trim(); // Trimmed: 'Hello World!'
```

2. Trim specific characters

You can pass a char array (or simply list characters) to remove specific leading and trailing characters:

```
string text = "###Important###";

string trimmedText = text.Trim('#');   // Trimmed: 'Important'
```

3. Trim from one end only

Use TrimStart() or TrimEnd() to remove characters from a specific side:

```
string str = "!!   Hello World!!";

string trimmedStart = str.TrimStart('!', ' ');   // Trimmed Start: 'Hello World!!'

string trimmedEnd = str.TrimEnd('!', ' ');   // Trimmed End: '!!   Hello World'
```

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ToLower() and ToUpper()

In C#, the methods for converting a string to lowercase or uppercase are ToLower() and ToUpper(), respectively. These are methods of the System.String class. They return a **new string**; they do not modify the original string instance.

These methods use the rules of the *current* computer's culture by default, which can lead to inconsistent behavior across different regional settings.

```
string original = "Hello, World!";

string upperCase = original.ToUpper(); // HELLO WORLD!

string lowerCase = original.ToLower();  // hello world!
```

H.W.

1. Write a program in C# Sharp to input a string and print it in reverse order.
2. Write a program in C# Sharp to compare two strings without using a string library functions.
3. Write a program in C# Sharp to count the number of alphabets, digits and special characters in a string.
4. Write a C# Sharp program to count the number of vowels or consonants in a string.
5. Write a C# Sharp program to read a sentence and replace lowercase characters with uppercase and vice-versa.
6. Write a program in C# Sharp to check the username and password.